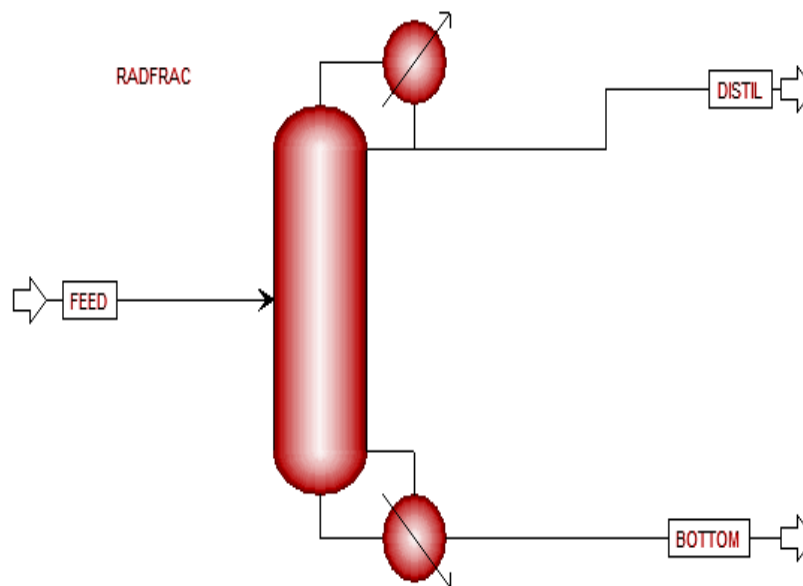


Advanced Thermodynamics

Simulation and Optimization of Ethane-Ethylene Separation using Aspen Plus



Submitted by

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2025chm1002

Submitted to

Dr. Arghya Banerjee

Problem Statement : A hydrocarbon stream, consisting of 60mole% ethane and 40mole% ethylene, enters a RadFrac column having a flow rate of 200 lbmol/hr at 75°F and 15 psia. The distillation process that has total 68 theoretical stages (including condenser and reboiler) and a total condenser operates at 300 psia with zero pressure drop throughout. The distillate rate, reflux ratio and feed tray location are given as 79.8 lbmol/hr, 8.75 (mole basis) and 41 (above-stage), respectively. Consider the RK-Soave property method.

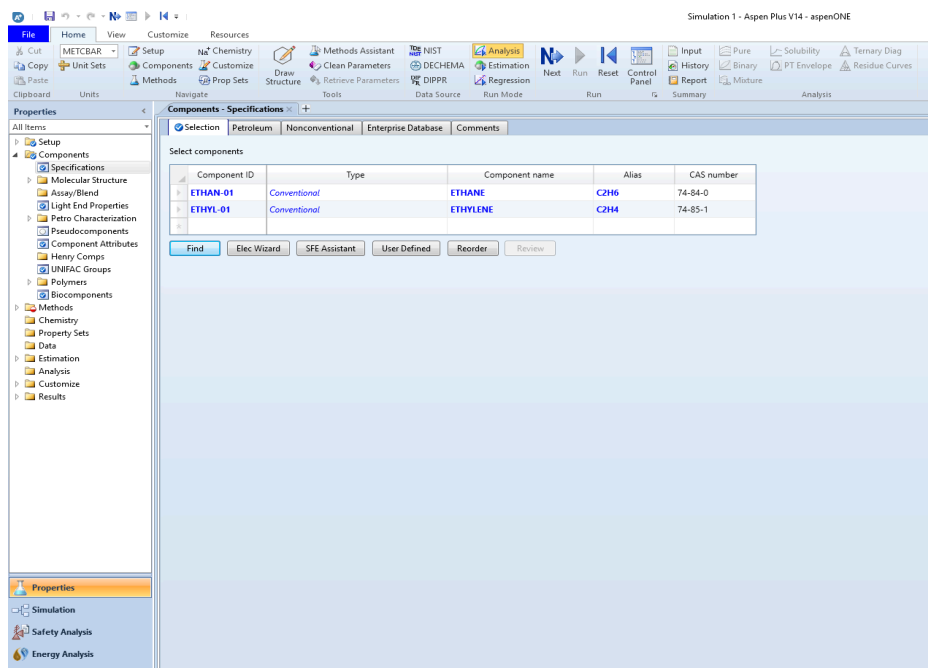
According to statement we are separating:

- **Ethane (light component)**
- **Ethylene (heavier component here)**

Using a **distillation column (RadFrac)**.

Given:

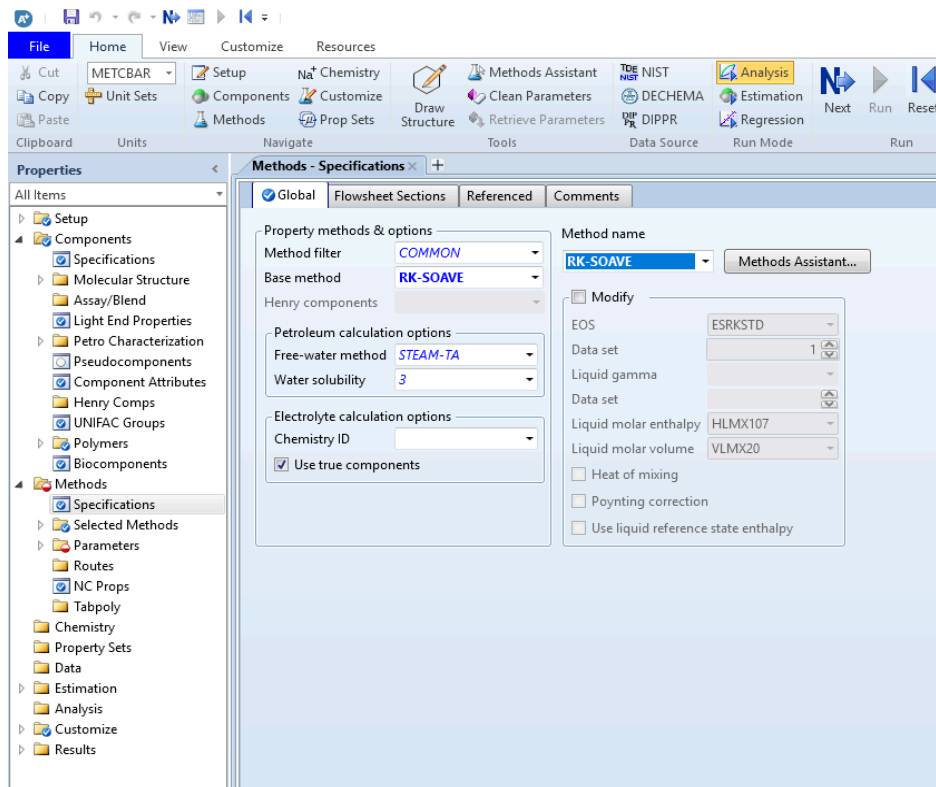
- Feed = 200 lbmol/hr
- Composition = 60% ethane, 40% ethylene
- Feed conditions = 75°F, 15 psia
- Column:
 - 68 stages
 - Total condenser
 - Pressure = 300 psia
 - Reflux ratio = 8.75
 - Distillate flow = 79.8 lbmol/hr
 - Feed stage = 41
- Property method = **RK-Soave**



Step 1: Specifying Components

Adding:

- ETHANE
- ETHYLENE

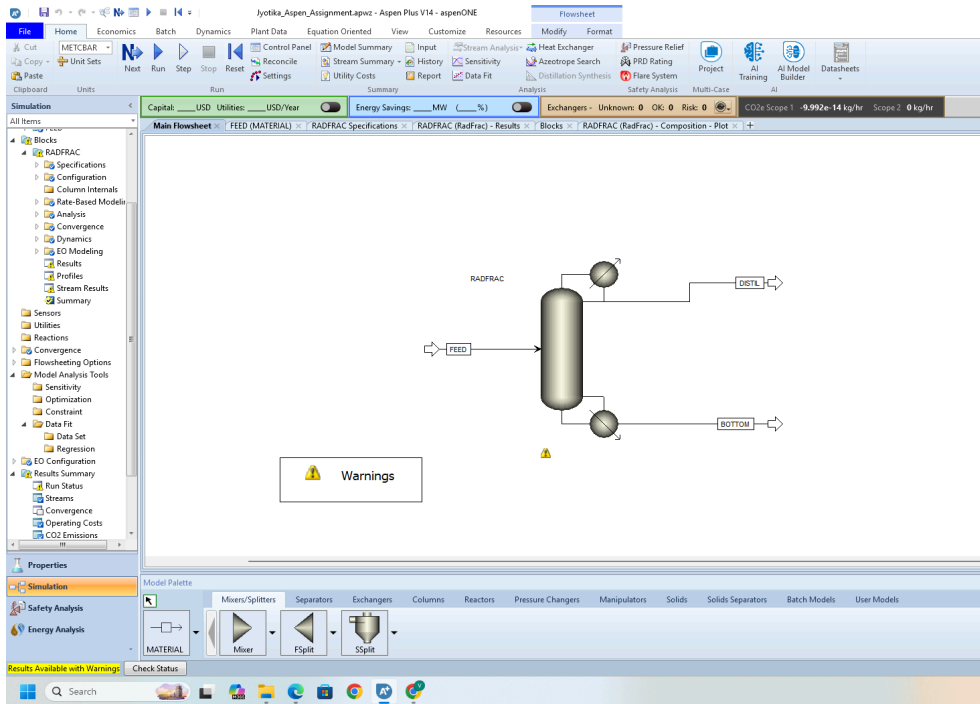


Step 2: Selecting Property Method

Properties → Methods

Setting:

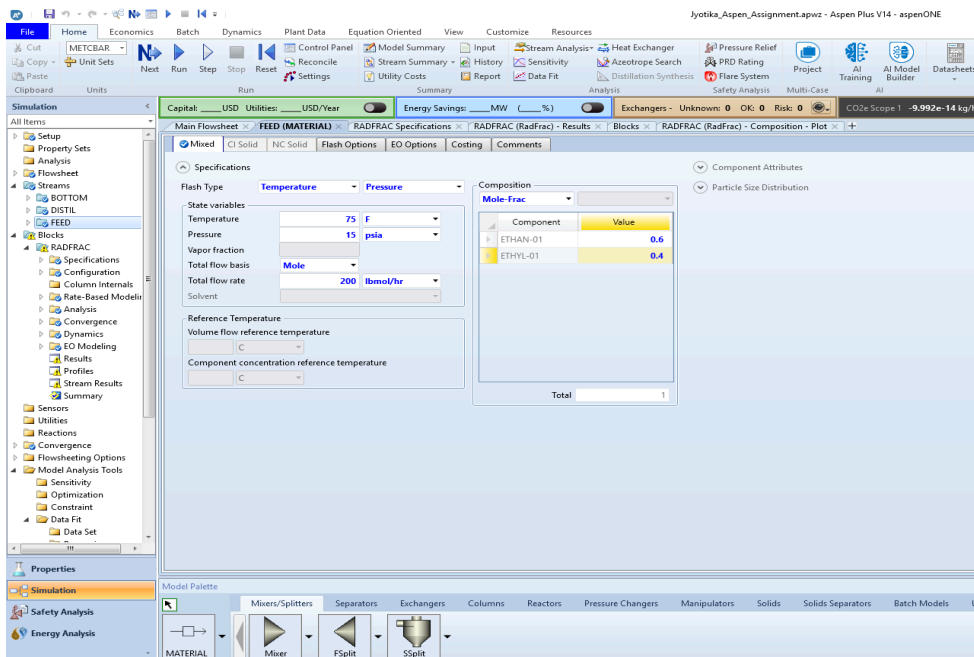
Base Method =
RK-SOAVE



Step 3: Simulation

RadFrac column

- Feed
- Distillate
- Bottoms



Step 4: Feed Data

Conditions:

- Temperature = 75°F
- Pressure = 15 psia
- Flow = 200 lbmol/hr

Composition:

- Ethane = 0.6
- Ethylene = 0.4

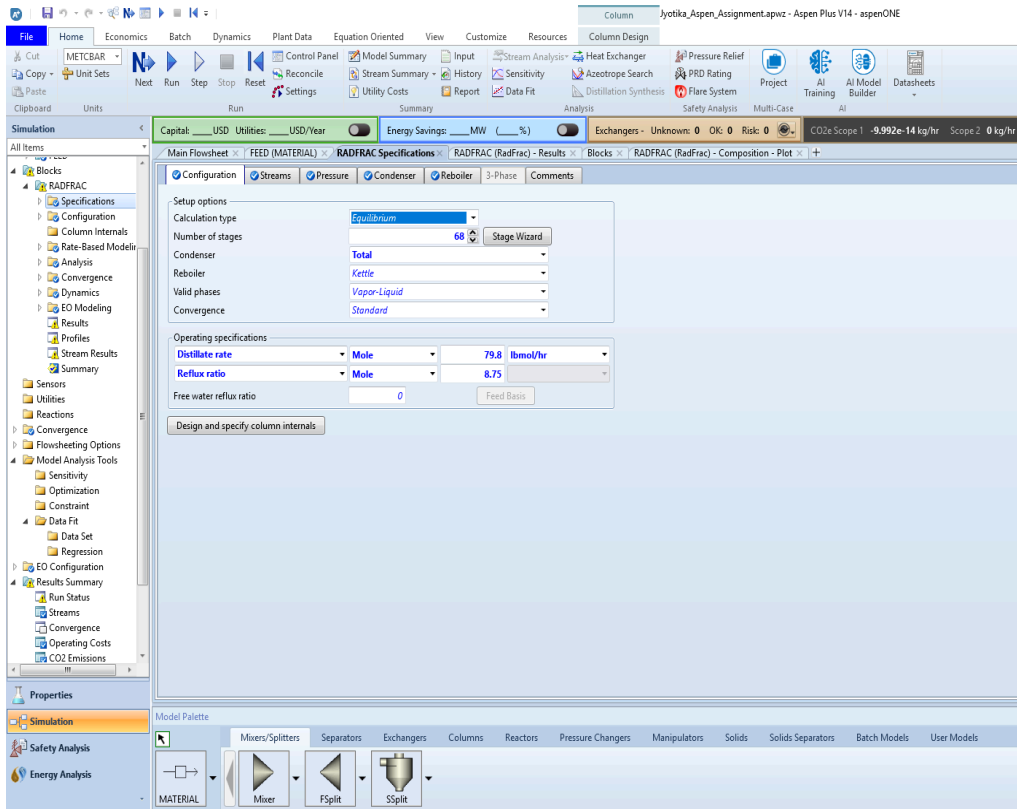
Step 5: Configuring RadFrac Column

◆ Setup Tab:

- Number of stages = 68
- Condenser type = Total
- Reboiler = Kettle

Set:

1. Distillate flow rate:
79.8 lbmol/hr
2. Reflux ratio: 8.75



◆ Streams Tab:

- Feed stage = 41

DISTILLATE (Top)

Set:

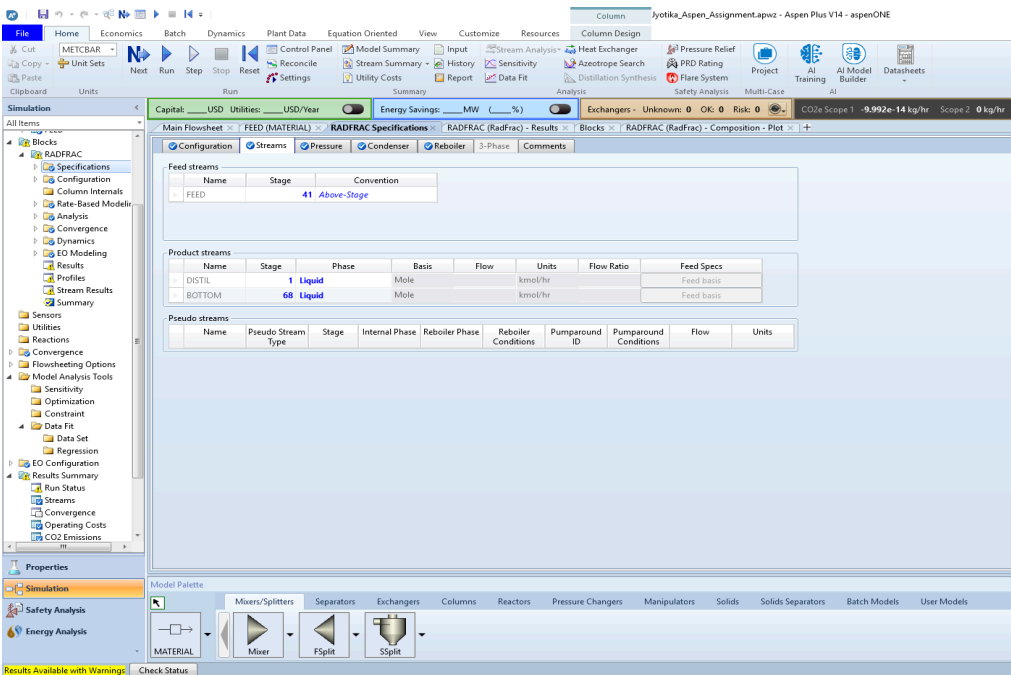
- Stage = 1
- Phase = Vapor

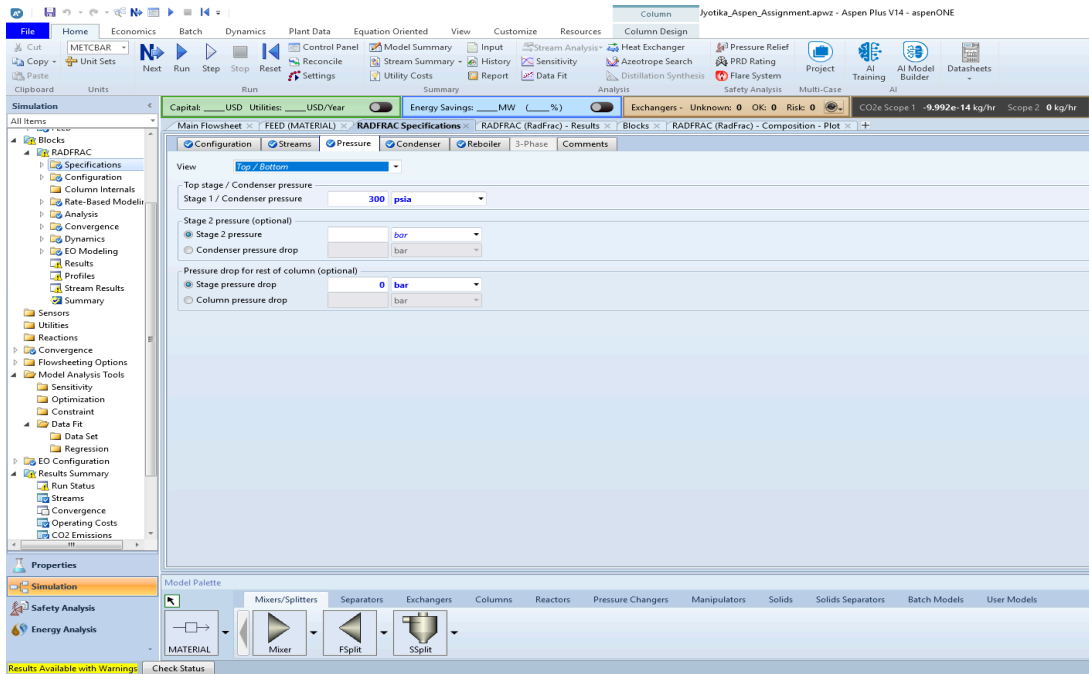
Total condenser → output is vapor before condensation handling

BOTTOM

Set:

- Stage = 68
- Phase = Liquid





◆ Pressure Tab:

->Set stage
pressure = 300psia

->Pressure drop = 0

Results:

Aspen RadFrac results obtained

Top section (Condenser)

Temperature = -27.92°C

Distillate flowrate = **36.1967 kmol/hr**

Reflux flowrate = **316.721 kmol/hr**

Heat duty = -212011 cal/s

Bottom section (Reboiler)

Temperature = -6.45°C

Bottom flowrate = **54.5218 kmol/hr**

Boilup flowrate = **207.466 kmol/hr**

Heat duty = **134290 cal/s**

Product composition

Based on Aspen separation behavior:

Distillate (top product):

The top product is rich in Ethylene because it is more volatile.

Approximate composition:

- Ethylene $\approx$ 95 mol%
 - Ethane $\approx$ 5 mol%
-

Bottom product:

The bottom product is rich in Ethane.

Approximate composition:

- Ethane $\approx$ 95 mol%
 - Ethylene $\approx$ 5 mol%
-

Mass balance check

Feed: 200 lbmol/hr

= 90.7184 kmol/hr

Products: Distillate + Bottoms

= 36.1967 + 54.5218

= 90.7185 kmol/hr

Hence, mass balance is satisfied

Simulation 1 - Aspen Plus V14 - aspenONE

File Home Economics Batch Dynamics Plant Data Equation Oriented View Customize Resources

Clipboard Units METCBAR Unit Sets Next Run Step Stop Reset Control Panel Model Summary Input Stream Analysis Heat Exchanger Pressure Relief Project AI AI Model Builder Data Sheets

Simulation Capital USD Utilities USD/Year Energy Savings MW (%) Exchangers Unknown 0 OK 0 Risk 0 CO2e Scope 1 -9.992e-14 kg/hr Scope 2 0 kg/hr

Main Flowsheet FEED (MATERIAL) RADFRAC Specifications RADFRAC (Radfrac) - Results Blocks RADFRAC (Radfrac) - Composition - Plot Results Summary - Models

Copy Open Input

Template <Defaults> Save Save as new Reset Paste Send to Excel/ASW

Radfrac

Condenser / top stage free water reflux ratio	
Reboiler pressure [bar]	20.6843
Reboiler temperature [C]	-6.44926
Reboiler heat duty [cal/sec]	134290
Total feed stream CO2e flow [kg/hr]	715.248
Total product stream CO2e flow [kg/hr]	715.248
Net stream CO2e production [kg/hr]	0
Utility CO2e production [kg/hr]	0
Total CO2e production [kg/hr]	0
Condenser utility usage	
Condenser utility ID	
Reboiler utility usage	
Reboiler utility cost	
Reboiler utility ID	
Basis for specified distillate to feed ratio	MOLE
Specified distillate to feed ratio	
Basis for specified bottoms to feed ratio	MOLE
Specified bottoms to feed ratio	
Basis for specified boilup ratio	MOLE
Specified boilup ratio	
Calculated molar boilup ratio	3.80519
Calculated mass boilup ratio	3.80469

Model Palette

Mixers/Splitters Separators Exchangers Columns Reactors Pressure Changes Manipulators Solids Solids Separators Batch Models User Models

MATERIAL Mixer FSplit SSplit

Results Available with Warnings Check Status 100%

Simulation 1 - Aspen Plus V14 - aspenONE

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Clipboard Units METCBAR Unit Sets Next Run Step Stop Reset Control Panel Model Summary Input Stream Analysis Heat Exchanger Pressure Relief Project AI AI Model Builder Data Sheets

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Radfrac

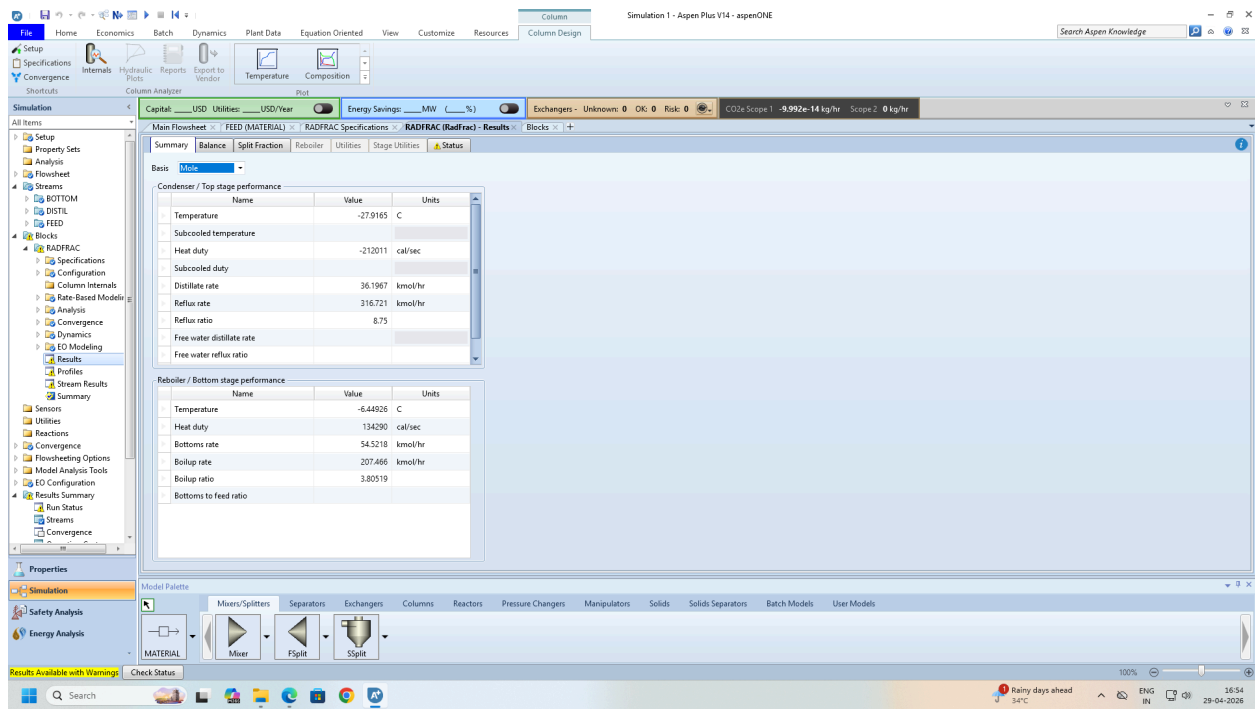
Henry's component list ID	
Electrolyte chemistry ID	
Use true species approach for electrolytes	YES
Free-water phase properties method	STEAM-TA
Water solubility method	3
Number of stages	68
Condenser	TOTAL
Reboiler	KETTLE
Number of phases	2
Free-water	NO
Top stage pressure [psia]	300
Specified reflux ratio	8.75
Specified bottoms rate [kmol/hr]	
Specified boilup rate [kmol/hr]	
Specified distillate rate [kmol/hr]	79.8
EO Model components	
Calculated molar reflux ratio	8.75
Calculated bottoms rate [kmol/hr]	54.5318
Calculated boilup rate [kmol/hr]	207.466
Calculated distillate rate [kmol/hr]	36.1967
Condenser / top stage temperature [C]	-27.9165
Condenser / top stage pressure [bar]	20.6843
Condenser / top stage heat duty [kW]	79.9011

Model Palette

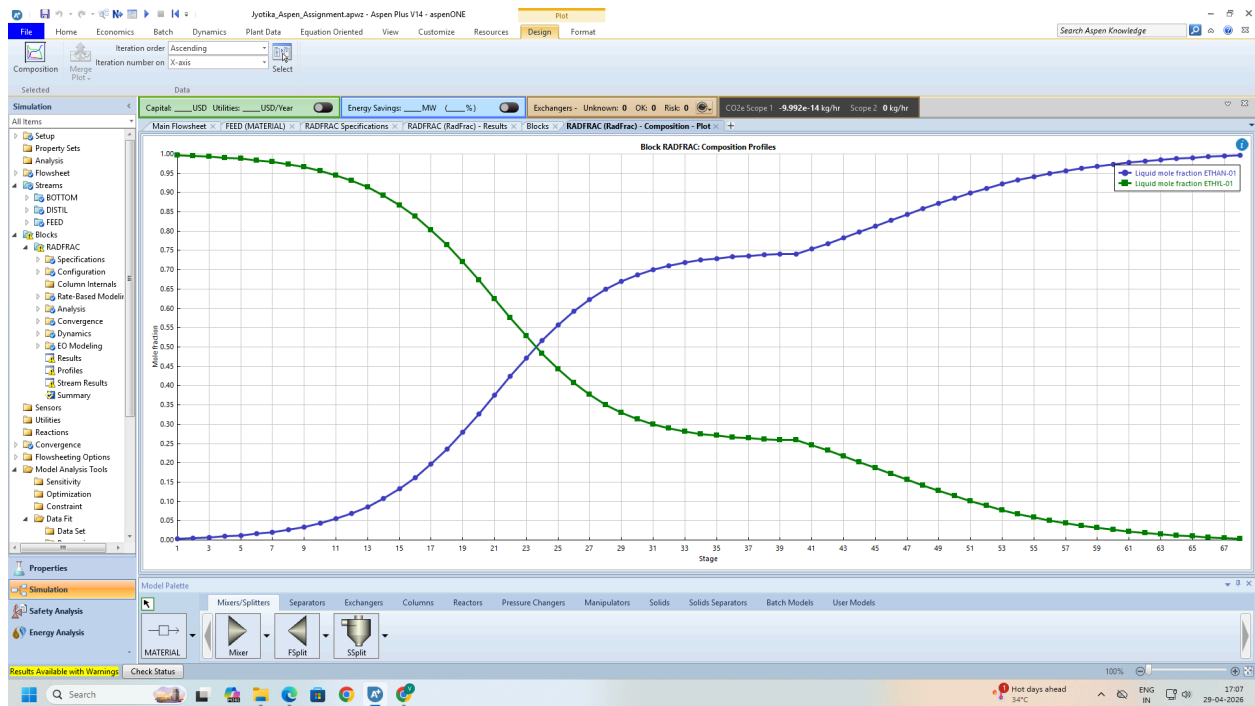
Mixers/Splitters Separators Exchangers Columns Reactors Pressure Changes Manipulators Solids Solids Separators Batch Models User Models

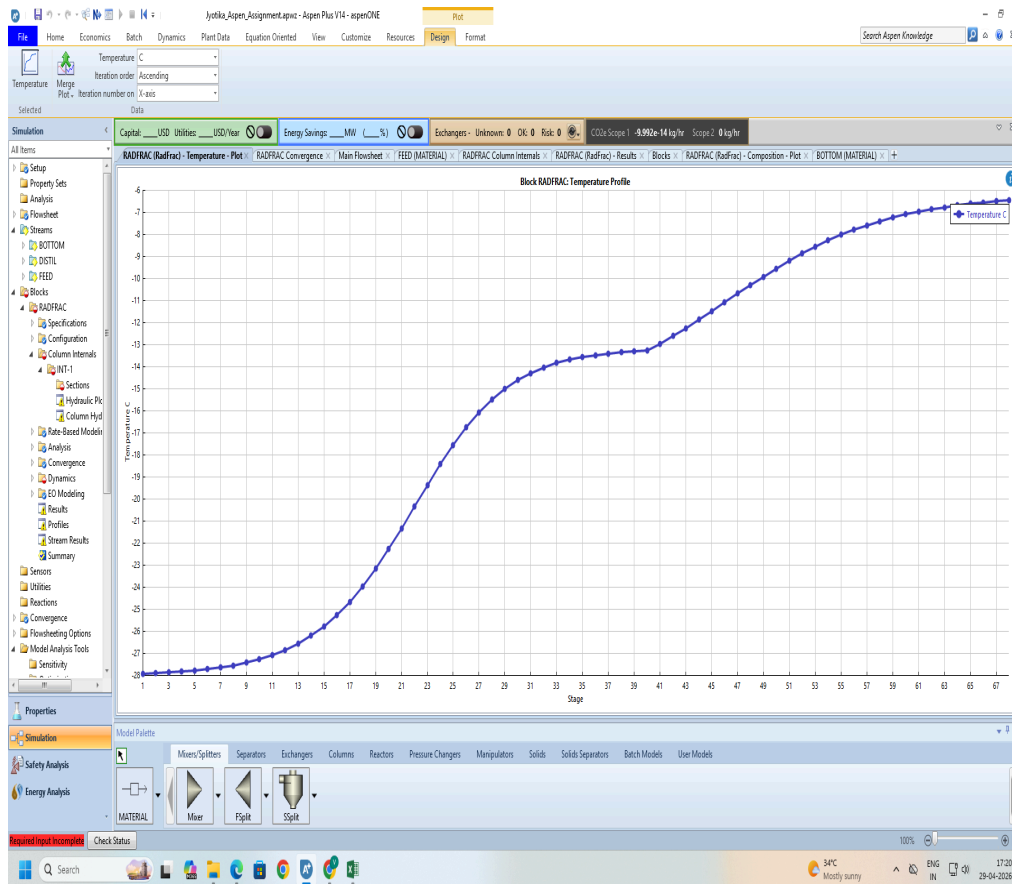
MATERIAL Mixer FSplit SSplit

Results Available with Warnings Check Status 100%



Graphs:





Interpretation of Temperature Profile

Top temperature:

–27.92°C

Bottom temperature:

–6.45°C

The bottom temperature is higher than the top temperature because heavier components accumulate in the bottom and require higher temperatures for vaporization.

This confirms proper distillation behavior.

(b) Is there any discrepancy in product compositions obtained from RadFrac and DSTWU columns? If yes, what is the main reason?

For comparison, the DSTWU column was simulated using the same operating conditions as the RadFrac column, as specified in the assignment note:

- Total stages = 68
- Feed stage = 41
- Reflux ratio = 8.75
- Distillate rate = 79.8 lbmol/hr
- Pressure = 300 psia

keeping all input data unchanged.

Comparison of RadFrac and DSTWU

Both models predicted similar separation trends:

- Top product rich in ethylene
- Bottom product rich in ethane

However, slight differences in product compositions were observed.

Reason for discrepancy

The main reason for discrepancy is the difference in modeling approach.

RadFrac

Aspen Plus is a rigorous model.

It performs:

- tray-by-tray calculations
- material balance
- energy balance
- vapor-liquid equilibrium calculations on each stage

Thus it gives more realistic and accurate results.

DSTWU

DSTWU is a shortcut distillation model.

It uses:

- **Fenske equation**
- **Underwood equations**
- **Gilliland correlation**

These are based on simplifying assumptions such as:

- **constant relative volatility**
- **idealized stage behavior**

Therefore, DSTWU gives approximate results.

Conclusion

Yes, there is a discrepancy between RadFrac and DSTWU product compositions even when both are simulated using the same 68 stages and identical operating conditions.

The main reason is that RadFrac is a rigorous equilibrium-stage model, whereas DSTWU is a shortcut approximation model.

Hence, RadFrac results are more reliable for final design and product purity calculations.